

Customer-Service Mirrored Avatar That Comes Beside You via Directional Spatial Audio

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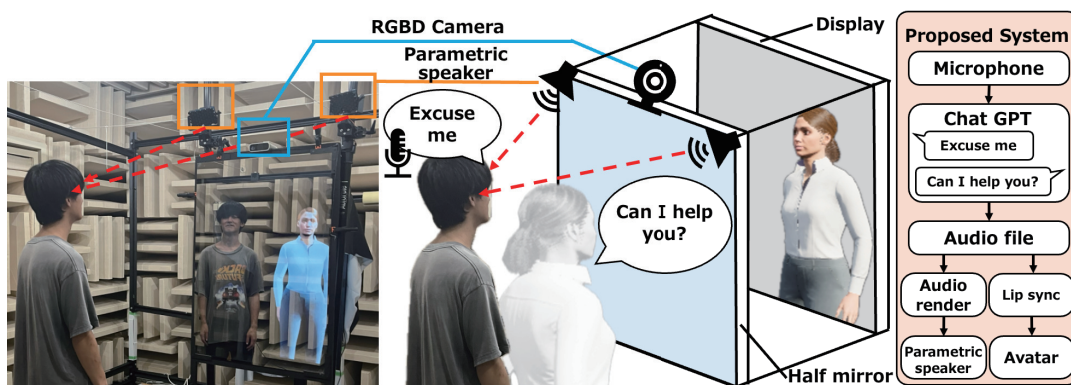


Figure 1: Overview of the proposed system: By integrating ChatGPT-driven interactive mirror avatars with a pair of user-tracking parametric speakers for precise 3D spatial audio, the system provides an experience as if the avatar were right beside the user, whispering.

Abstract

An interactive customer-service avatar system is proposed, integrating a half-mirror Augmented Reality (AR) display, user-tracking parametric array loudspeakers (PALs), and a ChatGPT dialogue module. The system delivers whisper-like, precise 3D audio selectively to the listener's ear. A preliminary study suggests this approach provides a selective and immersive acoustic experience for public-space customer service, potentially enhancing the user's sense of closeness and trust in the avatar.

Keywords

Mirrored Avatar, Directional Spatial Audio, Customer Service

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1 Introduction

Avatar-based customer service and guidance has been increasingly adopted owing to benefits such as workforce optimization and flexible operating hours [Kumar and Jhawar 2024]. At the same time, as exemplified by the “uncanny valley,” achieving visual and auditory realism remains challenging and critically shapes user's sense of affinity and trust. In particular, building positive rapport—such as a sense of closeness and trust—between customers and avatars is essential for improving customer satisfaction [Gremler and Gwinner 2000]. Prior work has proposed mirror-based avatar presentation methods that give the impression that the avatar actually exists in physical space [Uchida et al. 2021]. Compared with Virtual Reality (VR) or conventional display-based presentation, this approach has been shown to enhance the avatar's telepresence and overall user satisfaction; furthermore, because it does not require a head-mounted display, it enables immediate customer service in public spaces while preserving realism. However, these studies focus primarily on enhancing presence through visual presentation, with insufficient consideration of auditory realism. In particular, an “uncanny valley” also exists for sound-source localization: the perceived origin of the avatar's voice relative to the user strongly and

sensitively affects impressions of the avatar. Virtual reality experiments have shown that bringing the avatar’s auditory image closer to the user can improve rapport with the avatar [Zempo et al. 2022]. Therefore, combining mirror-based avatars with high-precision spatial audio that presents sound near the user irrespective of user position is expected to yield similar effects in real environments and to facilitate richer rapport formation.

Accordingly, we propose a high-precision spatial-audio avatar customer-service system that integrates a mirror-based avatar with PALs, whose presentation position is dynamically updated based on real-time estimation of the user’s position. PALs afford high-accuracy auditory localization, and their sharp directivity allows sound to be delivered directly to the user’s ears, creating an experience akin to being whispered to by a nearby avatar [Kuratomo et al. 2022]. Their directivity is also advantageous in public-space service scenarios from the perspectives of noise mitigation and privacy protection. To maintain naturalness in the interaction, we incorporate a pre-trained ChatGPT-based dialogue agent to ensure bidirectionality. This study examines whether, in mirror-based avatar scenarios, PAL-based audio can be perceived as emanating from the avatar’s virtual position near the user. Our goal is to demonstrate the feasibility of a real-time customer-service system that integrates vision and audition and fosters a heightened sense of closeness.

2 Proposed System

We propose an interactive customer-service system designed for high-realism, immersive avatar interactions in physical space, integrating a half-mirror AR display, spatial audio via PALs, and ChatGPT-based natural-language dialogue. A system schematic is shown in Fig. 1.

For visual presentation, the AR display renders virtual objects with depth in the mirror space. By inserting a gap, the apparatus provides natural stereopsis through binocular disparity and motion parallax, thereby enhancing perceived depth and presence. For acoustic presentation, we adopt sharply directive, real-time, user-tracking PALs to overcome conventional loudspeaker limitations, such as fixed sweet spots and localization degradation. A depth camera estimates the user’s 3D ear coordinates in real-time, allowing two servo-actuated PALs to track and steer sound delivery in azimuth and elevation for each ear. This setup enables high-precision, real-time sound localization irrespective of user position, delivering sound exclusively to the listener to create the impression of a nearby whisper. For dialogue, we integrate ChatGPT to realize an autonomous and natural conversational agent. User speech is captured, transcribed, and input to ChatGPT to generate a response. The generated text is converted to a WAV file via a text-to-speech engine. This audio is used to generate lip-sync animation for a MetaHuman avatar in Unreal Engine 5.

3 Experiment and Results

The objective of this experiment was to determine whether, in a mirror-avatar setting, acoustic presentation using PALs is perceived as originating closer to the user than presentation via conventional loudspeakers. As shown in Fig. 2, the study was conducted in an anechoic chamber with five participants. Speech synchronized with

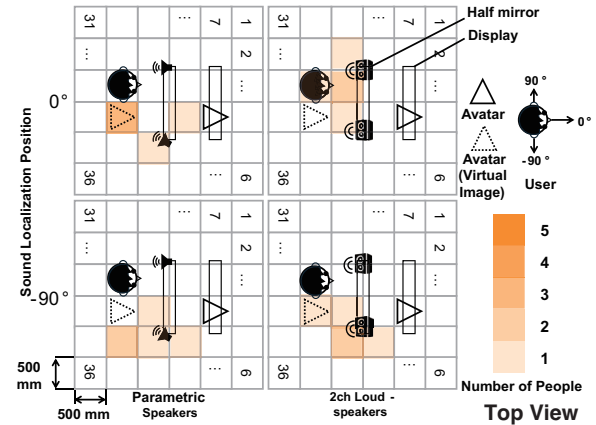


Figure 2: Experimental environment and results

the mirror avatar was spatialized and presented using either conventional two-channel loudspeakers or PALs. Stimuli were rendered from the participant’s right (-90°) and front (0°) azimuths. Participants reported perceived source locations by selecting indices on a numbered floor map on the ground (Fig. 2). The avatar executed a bow and lip-sync motion in synchrony with the audio.

As shown in Fig. 2, when the source was rendered at -90° , the PAL condition led more participants to perceive the sound accurately from the user’s right and on the near side of the mirror than did the loudspeaker condition. When rendered at 0° , loudspeaker responses clustered around the physical source, whereas under the PAL condition many participants localized the sound near the avatar’s virtual position on the near side of the mirror. This pattern is consistent with a ventriloquism effect whereby the right-side visual mirror avatar biases auditory localization; notably, the PAL’s ear-directed delivery likely strengthened the impression of being “whispered to.” These findings indicate that, relative to loudspeakers, the PAL-based system is more readily perceived as presenting sound from positions closer to the user, enabling an intimate, whisper-like experience with reduced dependence on listener position.

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